

Hospital Management Analysis (2024 - 2026)

Dashboard | Excel

Dataset Size

- **Total Records** : 1,000,000
- **Time Period** : 2024–2026
- **Coverage** : Multiple hospitals, cities and states
- **Data Type** : Hospital / Patient Visit Data
- **Primary Analysis Areas** : Patients, Doctors, Hospitals, Revenue, Appointments, Diagnoses, Payments and Visit Status

About the Dataset

- The dataset contains approximately 1,000,000 hospital records.
- The data represents hospital-related activities recorded during 2024 to 2026.
- It contains records from different hospitals located across different cities and states.
- Each record represents a patient visit/hospital transaction.
- The dataset contains information related to patients, doctors, hospitals, appointments, diagnoses, payments, and visits.
- It includes patient-related details such as patient ID, age, and gender.
- It includes doctor information, including unique Doctor IDs.
- It contains hospital and location information, allowing analysis across different hospitals, cities, and states.
- The dataset includes different medical procedures and diagnoses performed/recorded during patient visits.
- It contains appointment and visit information, including visit dates and visit status.
- Different payment methods are included, such as Cash, Card, UPI, Insurance, and Bank Transfer.
- The dataset also contains revenue-related information, which can be used to analyze hospital and city-wise revenue.
- Patient experience can be analyzed using satisfaction score, waiting time, and length of stay.
- The dataset contains both complete and repeated records, making it suitable for performing real-world data analysis and identifying duplicate records.
- Since the dataset contains a large number of records, it can be used to demonstrate large-scale data analysis in Excel.
- data disclaimer : This is a sample/synthetic dataset created for educational and portfolio purposes. It is designed to resemble real-world hospital data.

Data Processing and Cleaning

1. Duplicate Records Removal

- The dataset contained approximately 1,000 duplicate rows.
- Duplicate records were identified and removed while maintaining unique visit_id values.
- This helped avoid double-counting of patient visits during analysis.

2. Handling Missing Doctor IDs

- The doctor_id column contained several blank/missing values.
- Instead of assigning random or incorrect doctor IDs, the missing values were replaced with “Unknown”.
- This preserved the original information and clearly indicated that the doctor information was unavailable.

3. Standardizing Patient Gender Values

- The patient_gender column contained inconsistent text formats such as “Male”, “male”, “Female”, and “female”.
- These values were standardized into a consistent format (Male/Female) to avoid character and capitalization mismatches during analysis.

4. Handling Missing Patient Ages

- Some records had missing values in the original patient_age column.
- Instead of modifying the original age data, a separate column named patient_age_imputed was created.
- The average age calculated from the available patient-age values was 43 years.
- Missing age values were filled with 43 in the new patient_age_imputed column, while the original patient_age column was kept unchanged.
- This approach preserved the original dataset while creating a processed age field for analysis.

5. Creating Age Group

- A new column named age_group was created based on the patient_age_imputed column. Patients were categorized into the following groups:
- Infant: 1–11 years
- Child: 12 years. Teenager 13 - 18
- Adult: 19–63 years
- Senior: 64+ years

6. Removing Unrealistic Patient_Age

- Some records contained unusually high patient ages, such as 110, 113, 124, and 127 years.
- Since such ages are extremely rare and were considered unrealistic for the purpose of this dataset, these records were identified as potential data-entry errors and removed from the dataset.
- This helped maintain realistic patient-age values and improve the reliability of age-based analysis.

7. Handling Missing Payment Methods

- The payment_method column contained several blank/missing values.
- Instead of assuming or assigning a specific payment method, the missing values were replaced with “Unknown”.
- This preserved the records while clearly indicating that the payment method information was unavailable.

8. Handling Missing Visit_Status

- Some records contained blank values in the visit_status column.
- These missing values were replaced with “No Show” to maintain consistency in the visit-status field and ensure that missing entries did not create issues during analysis.

9. Handling Missing Waiting Time

- The waiting_time_minutes column contained some blank values.
- The average waiting time calculated from the available records was 34 minutes.
- The missing values were therefore replaced with 34 minutes, allowing the column to be used consistently for waiting-time analysis without removing the affected records.

10. Handling Missing Length_of_Stay

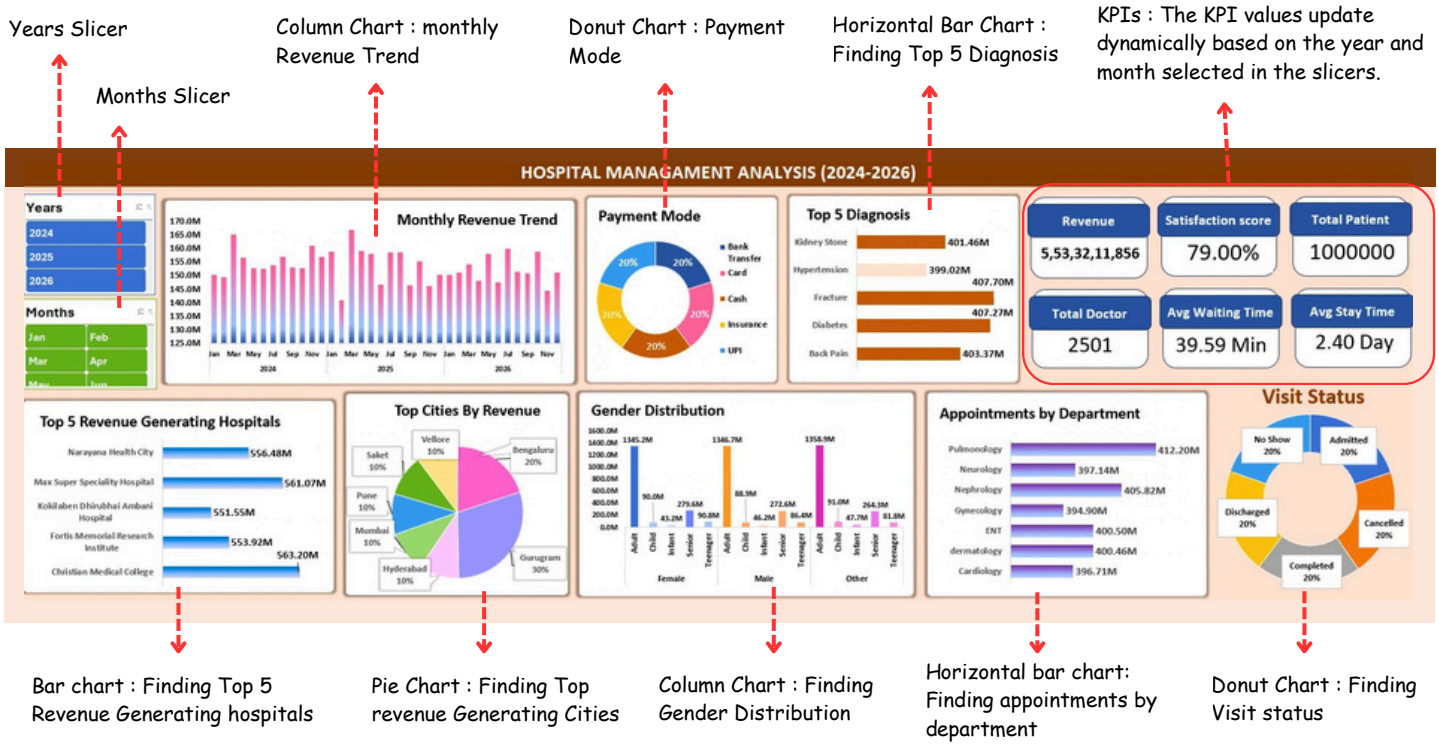
- Some records contained missing values in the length_of_stay column.
- The average length of stay calculated from the available records was 2 days.
- The missing values were replaced with 2 days to maintain a complete dataset for length-of-stay analysis.

11. Handling Missing Satisfaction_Score

- The satisfaction_score column contained some blank values.
- The average satisfaction score calculated from the available records was 3.4.

- The missing values were replaced with 3.4, allowing satisfaction-related analysis to be performed without excluding records with missing scores

Interactive Dashboard Understanding



The dashboard is fully interactive; selecting specific years or months in the slicers automatically updates all KPIs, charts, and graphs to help easily extract specific data

Key Insights

1. Analyzing the overall performance of the healthcare network from **2024 to 2026**, encompassing total revenue generation, patient demographics, top-performing hospitals, and department-wise contributions to summarize key business insights

Total Revenue : ₹ **5533211856** (which is around 553 Cr)

Total Patients : **1000000** (which is 10 lakh)

Total Distinct Doctors : **2500**

Average Patient Satisfaction Score : **79.00%** (as per data ratings given by 5 so approx 4/5)

Average Waiting Time Before Checkup : (**39.59 Min** which is approx 40 Mins)

Average Stay Time : **2.40 Day** which is approx Two n half day

Top 5 Revenue Generated Hospitals ---- (Hospital's Name) --- (State) --- (City) --- (Revenue)

1. Christian Medical College -----	Tamil Nadu -- Vellore -----	₹ 563203374.8 (approx 563.20M)
2. Max Super Speciality Hospital -----	Delhi ----- Saket -----	₹ 561073059.0 (approx 561.07M)
3. Narayana Health City hospital -----	Karnataka --- Bengaluru --	₹ 556476761.9 (approx 556.48M)
4. Fortis Memorial Research institute -----	Hararyana ---- Gurugram --	₹ 553919531.0 (approx 553.92M)
5. Kokilaben dhirubhai Ambani hospital -	Maharashtra -Mumbai ---	₹ 551549868.4 (approx 551.55M)

Top 5 Diagnosis and revenue generated on it.

1. Fracture	: ₹ 407704154.1 (approx 407.70M)
2. Diabetes.	: ₹ 407271456.4 (approx 407.27M)
3. Back Pain.	: ₹ 403374601.6 (approx 403.37M)
4. kidney stone	: ₹ 401456771.2 (approx 401.46M)
5. Hypertension	: ₹ 399021150.6 (approx 399.02M)

Top 5 Revenue Generated Cities (2024 - 2026 Total)

1. Gurugram	: ₹ 1644677293 (which is around 30%)
2. Bengaluru	: ₹ 1105441509 (which is around 20%)
3. Vellore.	: ₹ 566357930.2 (which is around 10%)
4. Saket.	: ₹ 563895384.1 (which is around 10%)
5. Mumbai	: ₹ 554766151.4 (which is around 10%)
6. Pune.	: ₹ 552498635.6 (which is around 10%)
7. Hyderabad	: ₹ 545574952.6 (which is around 10%)

Top 5 Departments and their revenues (2024 - 2026 Total)

1. Pulmonology	: ₹ 412201711.5 (approx 412.20M)
2. Nephrology	: ₹ 405817860.8 (approx 405.82M)
3. ENT.	: ₹ 400499872.8 (approx 400.50M)
4. Dermatology.	: ₹ 400463696.7 (approx 400.46M)
5. Neurology.	: ₹ 397137774.6 (approx 397.14M)

Gender Revenue Distribution (2024 - 2026 Total)

		2024	2025	2026	Total
FEMALE	Adult	₹ 449093142.1	₹ 447531129	₹ 448574419.5	₹ 1345198691
	Child	₹ 29997536.72	₹ 28052822.03	₹ 31927436.5	₹ 89977795.25
	Infant	₹ 13621265.65	₹ 14031781.45	₹ 15504685.03	₹ 43157732.13
	Senior	₹ 101579646.9	₹ 86892802	₹ 91159873.22	₹ 279632322.1
	Teenager	₹ 31737428.69	₹ 30950085.76	₹ 28085399.2	₹ 90772913.65
MALE	Adult	₹ 437348953.3	₹ 463558374.2	₹ 445821191.6	₹ 1346728519
	Child	₹ 30804465.48	₹ 29998375.13	₹ 28062868.69	₹ 88865709.3
	Infant	₹ 13612086.63	₹ 13780536.5	₹ 18838208.07	₹ 46230831.2
	Senior	₹ 94461330.71	₹ 90968891.32	₹ 87151527.29	₹ 272581749.3
	Teenager	₹ 28473203.55	₹ 27730402.67	₹ 30222041.81	₹ 86425648.03
OTHERS	Adult	₹ 471898668.3	₹ 447330069.6	₹ 439664596.2	₹ 1358893334
	Child	₹ 29318694.17	₹ 31315543.48	₹ 30359334.17	₹ 90993571.82
	Infant	₹ 14324503.46	₹ 17366798.69	₹ 15983193.19	₹ 47674495.34
	Senior	₹ 87369234.21	₹ 89888950.17	₹ 87004106.79	₹ 264262291.2
	Teenager	₹ 27858930.09	₹ 26174796.18	₹ 27782526.29	₹ 81816252.56
Total		₹ 1861499090	₹ 1845571358	₹ 1826141408	₹ 5533211856

2. Demographic Insight

Dominant Patient Group :

Overall, the Adult category (ages 19–63) recorded the highest patient volume under the "Others" classification which is ₹ 1358893334 , indicating that working-age adults form the primary user base.

Trend Shift in Minimum Volumes:

In 2024 and 2025, the Infant male demographic consistently recorded the lowest patient count. In contrast, the trend shifted in 2026, with the Infant female category registering the lowest volume.

Conclusion :

The analysis highlights that adults (ages 19–63) remain the core demographic utilizing healthcare services. Furthermore, the shift in minimum patient volumes—from infant males in previous years to infant females in 2026—points to changing pediatric admission patterns that require closer monitoring for child healthcare planning.

3. Payment Mode Trends & Digital Adoption Insight

Consistent Overall Distribution:

Across the entire timeframe, payment methods show a well-balanced distribution where Cash, Bank Transfer, Card, Insurance, and UPI each account for approximately 20% overall.

Year-on-Year Trend (2024–2025):

In both 2024 and 2025, individual payment modes maintained an even 20% share uniformly.

Shift in 2026 (Digital Adoption):

By 2026, a slight behavioral shift occurred where card transactions decreased marginally to 19%, while UPI payments increased to 21% (with remaining modes staying at 20%).

Conclusion :

The steady distribution of payment methods highlights a diversified approach to healthcare transactions. Furthermore, the shift observed in 2026—marked by a decrease in card usage and a rise in UPI payments—reflects a growing trend toward digital transformation and technological adoption among patients

4. Top Highest Revenue Generated Hospital Each Year

2024 (Format -----(Hospital's Name) ----- (State) ----- (City) ----- (Revenue)

- 1.Forties Memorial Research Institute ---- Haryana ----- Gurugram --- ₹193231863.1
- 2.Narayana Health City ----- Karnataka ----- Bengaluru ---- ₹ 191985352.6
- 3.Columbia Asia Hospital ----- Maharashtra ---- Pune ----- ₹ 191615745.3

2025

- 1.Narayana Health City ----- Karnataka ----- Bengaluru ---- ₹ 192060277.5
- 2.Manipal Hospital ----- Karnataka----- Bengaluru----- ₹ 186844417.7
- 3.Christian Medical College ----- Tamil Nadu ---- Vellore ----- ₹ 186072770.1

2026

- 1.Christian Medical College ----- Tamil Nadu ----- Vellore ----- ₹195455302.4
- 2.Max Super speciality hospital ----- Delhi ----- Saket ----- ₹ 189204628.6
- 3.Columbia Asia Hospital ----- Maharashtra ----- Mumbai ----- ₹ 186555558.2

Performance Shift in Bengaluru Hospitals (Narayana & Manipal):
During 2024 and 2025, Bengaluru-based hospitals (particularly Narayana Health City) demonstrated exceptional financial performance and generated massive revenue, showing high stability. However, by 2026, no hospital from Bengaluru managed to secure a spot in the top 3 rankings.

Potential Cause & Investigation:
This sudden drop—taking Narayana out of even the top 4 after years of stability—requires immediate investigation. It is critical to inspect operational bottlenecks, such as potential equipment or machinery malfunctions/downtime, that may have impacted patient capacity and revenue generation.

Rise of Christian Medical College (CMC), Vellore (Tamil Nadu):
In stark contrast, Christian Medical College (CMC), Vellore (Tamil Nadu) has shown stellar and consistent performance throughout 2025 and 2026, emerging as a top-performing healthcare institution.

Conclusion :

The significant performance volatility observed in premier institutions like Narayana Health City highlights the vulnerability of high-volume centers to operational disruptions. To sustain leadership and reverse such declines, management must conduct a thorough internal audit of advanced medical machinery and diagnostic equipment.
Furthermore, building on the success seen in institutions like CMC Vellore, the network should focus on expanding specialized healthcare services, ensuring zero stock-outs of essential medicines, and maintaining optimal availability of expert medical professionals to guarantee uninterrupted, high-quality patient care.

5. Sudden Shift in disease volume :

In 2024, Diabetes recorded the highest case volume across the network, reaching approximately 149.71M. However, it abruptly dropped out of the top 5 categories in 2025. Meanwhile, in 2026, Fracture cases emerged as the highest volume at approximately 143.95M.

Analytical Insight & Interpretation:

From a clinical perspective, a chronic condition like diabetes does not act as a seasonal flu; it cannot realistically experience such a massive, sudden drop overnight. Therefore, this drastic variation strongly points toward data collection artifacts, changes in diagnostic coding/classification, or categorization shifts across years rather than a real-world shift in patient demographics. This highlights the need for standardized data entry and consistent disease coding protocols in future data tracking

6. Regional Performance Analysis: Low-Revenue Insights (Delhi & Hyderabad)

Observation on Low-Performing Centers:

Throughout the analysis period, hospitals in major metropolitan hubs—specifically Max Super Speciality Hospital (Saket, Delhi) and Apollo Hospital (Hyderabad, Telangana)—consistently recorded the lowest revenue generation about (56Cr and 54Cr) across the network.

Strategic Interpretation & Recommendations:

While hospitals in regions like Delhi and Hyderabad operate in highly competitive healthcare markets, their lower revenue performance compared to other tier regions warrants a dedicated operational review. Management should investigate underlying factors such as regional pricing models, local market competition, patient volume acquisition, and cost structures in these specific branches to improve their financial viability and market penetration."

7. Year-on-Year Revenue & Patient Volume Trend Analysis (2024–2026)

Quantitative Decline Overview:

2024

Generated a Total Revenue of ----- ₹ 1,861,499,090 with ----- 334,020 patients.

2025

Recorded a slight dip with a revenue of ----- ₹ 1,845,571,358 and ----- 333,099 patients.

2026

Further decreased to a revenue of ----- ₹ 1,826,141,408 with ----- 332,881 patients.

As observed from the data above, there is a consistent pattern of a gradual decline in both overall revenue and patient volume over the years."

Trend Insight & Retention Analysis:

A steady year-on-year decline is observed in both overall revenue and total patient volume from 2024 through 2026. This downward trend suggests a potential challenge in patient retention—indicating that a significant number of patients who visit once may not be returning for follow-up care. This pattern points toward possible gaps in patient satisfaction, service quality, or post-treatment engagement.

Strategic Recommendation:

To reverse this decline, hospital management must prioritize enhancing the overall quality of care. Implementing systematic patient feedback mechanisms, improving post-visit follow-up communications, and ensuring a seamless, high-quality patient experience at every touchpoint will be critical to boosting patient loyalty, retention, and long-term financial growth.

8. Strategic Recommendations & Future Action Plan

Machine Availability for High-Revenue Departments:

Given that Pulmonology stands as the highest revenue-generating department, the hospital network must ensure that all specialized respiratory and diagnostic machinery is fully functional, well-maintained, and sufficiently available to support continuous patient care.

Inventory Readiness for High-Frequency Cases:

Since fracture cases record the highest overall volume across the network, management must maintain robust stock levels of orthopedic supplies, casts, and emergency equipment to avoid any treatment delays.

Patient Wait Time Optimization:

Implement an advanced appointment scheduling and digital queuing system to streamline patient registration and reduce average waiting times.

Encouraging Digital Transactions:

Promote cashless and mobile-based payment methods (such as UPI) at all billing counters to build upon the growing digital adoption trend observed in 2026.

Proactive Inventory & Pharmacy Management:

Deploy an automated inventory tracking system to ensure 24/7 availability of essential medicines and completely eliminate stock-out situations.

Optimized Resource & Staff Allocation:

Plan shift-wise doctor and medical staff availability strategically to handle peak-hour patient loads efficiently and maintain high-quality care.

